Finite Phase Invariants of Normalized Weighted Matrices: Exact Computation, Perturbation Certificates, and Causal Detection

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1. Introduction

Many evolving systems are summarized at each instant by a weighted directed graph, that is, by a nonnegative matrix whose entries record the magnitude of a relation between two nodes. A recurring problem is to distinguish a *discrete structural transition* in such a graph — a qualitative change in how the nodes are connected — from a merely *continuous change in magnitude*. The two are different in kind: the first is a change of class, the second a change of size, and a statistic that conflates them is of little use. This paper constructs an integer-valued invariant of a normalized nonnegative matrix that is insensitive to smooth deformation, that is computable exactly, and that is accompanied by a stability certificate computed from observed data alone; and it separates this invariant cleanly from every estimate of magnitude.

The invariant is obtained by embedding the normalized matrix into the link variables of a finite periodic lattice gauge field and reading an integer phase C from the plaquette holonomies. This C is a discrete first Chern number in the sense made precise for the quantized Hall conductance by Thouless, Kohmoto, Nightingale, and den Nijs [2] and for adiabatic phases by Berry [3]; it is computed here by the lattice procedure of Fukui, Hatsugai, and Suzuki [5], and is realized concretely by two-band constructions of the type introduced by Qi, Wu, and Zhang [6]. Its continuum antecedent is the Chern–Simons functional of Chern and Simons [1], whose role in knot and link invariants was established by Witten [4]. We record that functional only as background (Lemma 14); none of the finite results depends on it.

Two properties make the invariant usable. First, C is computed in exact rational arithmetic, with no evaluation of π and no trigonometric functions (Lemma 4), so there is no rounding ambiguity in the integer it returns. Second, a certificate computable from observed data alone (Theorem 6) guarantees that C is constant over an entire ℓ^1-ball of admissible inputs; measurement error is thereby controlled without reference to an unknown true phase.

Because the construction uses no metric, it carries no magnitude, and we make this precise rather than leave it implicit. A dimensionless invariant composed with a positively homogeneous functional returns only that functional's own scale (Lemma 7), and no function of the invariant alone can recover magnitude (Proposition 8); specialized to loss, this concerns quantiles and the Expected Shortfall whose coherence and representation

were studied by Artzner, Delbaen, Eber, and Heath [7], by Rockafellar and Uryasev [8], and by Acerbi and Tasche [9]. A conditional-expectation identity (Proposition 10), with conditional expectation read as the L^2 projection in the sense of Williams [10], gives the exact condition under which adjoining C to other predictors reduces mean-squared error — and shows that when C is a deterministic function of the full inputs it yields no reduction whatsoever. The invariant's contribution is therefore delimited to the finite-sample or computational, never the informational.

Finally, a finite-state automaton (Definition 4) converts a stream of certified labels into a causal detector with proven safety and reproducibility (Theorem 11), a bound on spurious activations (Proposition 12), and a detection-delay bound (Proposition 13). A quantum-channel marginal-invariance lemma (Lemma 15) — the no-signalling property argued by Ghirardi, Rimini, and Weber [11] and formalized in the channel language of Nielsen and Chuang [12] — shows that a distributed read-out cannot shortcut the arrival of information.

Section 2 fixes notation and the three constructions used throughout. Section 3 proves the normalization and embedding bounds; Section 4 constructs the integer invariant and its exact computation; Section 5 proves the perturbation certificate. Section 6 establishes scale separation and non-identifiability, and Section 7 the conditional-expectation identity. Section 8 defines the detection automaton and proves its guarantees. Section 9 records the continuum functional as background, and Section 10 the distributed-read-out invariance.

To facilitate reproducibility, Python validation scripts accompany the mathematical results identified in Finite Phase Invariants of Normalized Weighted Matrices: Exact Computation, Perturbation Certificates, and Causal Detection. The scripts implement the described algorithms and perform exact finite computations and numerical consistency checks supporting the analysis. These computational checks complement, but do not replace, the mathematical proofs and their stated assumptions. The code is available at:

https://github.com/teknoquants/Finite-Phase-Invariants-of-Normalized-Weighted-Matrices

2. Setting and definitions

Definition 1. Let $N \geq 2$ and let $W = (W_{ij})$ be a nonnegative $N \times N$ matrix with $W_{ii} = 0$ and total mass $S = \sum_{ij} W_{ij} > 0$. Write $\| M \|_1 = \sum_{ij} |M_{ij}|$ and $X = W/S$. Let $\widehat{W}$ be a second such matrix with mass $\hat{S}$ and $\hat{X} = \widehat{W}/\hat{S}$, and let $\varepsilon > 0$ satisfy $\| W - \widehat{W} \|_1 \leq \varepsilon < \hat{S}$.

Definition 2. Fix an integer $n \geq 2$ with $r := N(N-1) \leq 4n^2$. On the $n \times n$ periodic lattice (indices mod n) assign to each site (i, j) two complex link variables $z_x(i, j), z_y(i, j)$, giving $4n^2$ real coordinates. Fix nonzero rational baseline links z^0, a rational $\kappa > 0$, and an injection J of the r off-diagonal entries of X (listed in a fixed order as $x \in \mathbb{R}^r$) into distinct real link coordinates. Set $z(X) = z^0 + \kappa J x$.

Definition 3. For links with $z_e \neq 0$ set $U_e = z_e/|z_e|$, and for each site

$$R_{ij} = z_x(i,j)\, z_y(i+1,j)\, z_x(i,j+1)^*\, z_y(i,j)^*, \qquad P_{ij} = R_{ij}/\big|R_{ij}\big|.$$

Call a site *admissible* if $R_{ij} \notin (-\infty, 0]$. When all sites are admissible, with $\mathrm{Arg} \in (-\pi, \pi)$,

$$C = \frac{1}{2\pi} \sum_{ij} \mathrm{Arg}\, R_{ij}.$$

Definition 4. Fix $K \subset \mathbb{Z}$, positive integers h_B, h_A, h_R, and $D \geq 0$. Each input is of type S (an admissible label $c \notin K$), type C (an admissible label $c \in K$), or $\perp$. The states are BOOT, READY, ARMED; a counter of consecutive like inputs is reset by any contrary or $\perp$ input. Transitions: BOOT → READY after h_B consecutive S; READY → ARMED after h_A consecutive C (recording the time); ARMED → READY after h_R consecutive S with elapsed time $\geq D$; any $\perp$ resets counters and sends BOOT, READY → BOOT while fixing ARMED. Outputs are 0 in BOOT, READY, 1 in ARMED, and $\perp$ on an invalid input. Decision times strictly increase.

3. Normalization and embedding

Lemma 1. With Definition 1, $X(sW) = X(W)$ for $s > 0$, and $\| X - \hat{X} \|_1 \leq 2\varepsilon/\hat{S}$.

Proof. Nonnegativity gives $\| W \|_1 = S$. Since $\big|S - \hat{S}\big| \leq \| W - \widehat{W} \|_1 \leq \varepsilon < \hat{S}, S > 0$. Write $X - \hat{X} = \big(W - \widehat{W}\big)/\hat{S} + W\big(1/S - 1/\hat{S}\big)$. The first term has $\|\cdot\|_1 \leq \varepsilon/\hat{S}$; the second has $\|\cdot\|_1 = \| W \|_1 \ \big|S - \hat{S}\big|/\big(S\hat{S}\big) = \big|S - \hat{S}\big|/\hat{S} \leq \varepsilon/\hat{S}$. Their sum is $\leq 2\varepsilon/\hat{S}$. The scale identity follows by cancellation of s. ■

Lemma 2. With Definition 2, each X_{ij} is recovered from its coordinate by subtracting z^0 and dividing by κ, so J is injective, and

$$\max_e \big|z_e(X) - z_e\big(\hat{X}\big)\big| \ \leq \ \kappa \ \| X - \hat{X} \|_1 \leq \ \frac{2\kappa\varepsilon}{\hat{S}} \ =: \ d.$$

Proof. A link of $\kappa J x$ differs from the corresponding link of $\kappa J \hat{x}$ in at most its two real coordinates; its modulus change is at most the sum of those coordinate changes, which is at most $\kappa \| x - \hat{x} \|_1 = \kappa \| X - \hat{X} \|_1$. Apply Lemma 1. ■

4. The integer invariant and its exact computation

Lemma 3. If all sites are admissible, then $C \in \mathbb{Z}$ and $|C| < n^2/2$. C is invariant under $U_\mu(v) \mapsto h(v)^* U_\mu(v) h\big(v + e_\mu\big)$ with $|h(v)| = 1$; conjugating every link sends $C \mapsto -C$; and C is constant along any continuous path of nonvanishing links on which no plaquette enters $(-\infty, 0]$.

Proof. In $\prod_{ij} P_{ij}$ each link occurs once unconjugated and once conjugated by periodicity, and $|U_e| = 1$, so $\prod_{ij} P_{ij} = 1$ and $\exp\big(i \sum_{ij} \mathrm{Arg}\, R_{ij}\big) = 1$; hence $\sum_{ij} \mathrm{Arg}\, R_{ij} \in 2\pi\mathbb{Z}$ and $C \in \mathbb{Z}$.

Each admissible term has $|\mathrm{Arg} R_{ij}| < \pi$ and there are n^2 terms, so $|C| < n^2/2$. The vertex factors cancel within each plaquette. Conjugation negates each argument. Along an admissible path $\sum_{ij} \mathrm{Arg}\, R_{ij}$ is continuous and valued in $2\pi\mathbb{Z}$, hence constant. ■

Lemma 4. If all links are rational complex numbers, C is computable by integer arithmetic and rational sign tests, with no evaluation of π or of trigonometric functions.

Proof. Let $\mathrm{side}(q) = +1$ if $\mathrm{Im} q > 0$ or q is a negative real, -1 if $\mathrm{Im} q < 0$, and 0 if q is a positive real. Initialise $a = 1, w = 0$; processing each admissible R in a fixed order, set $v = aR$, add 1 to w if $\mathrm{side}(a) = \mathrm{side}(R) = +1$ and $\mathrm{side}(v) \le 0$, subtract 1 if $\mathrm{side}(a) = \mathrm{side}(R) = -1$ and $\mathrm{side}(v) \ge 0$, and replace a by v (optionally divided by any common positive factor). The invariant $\sum_{\text{processed}} \mathrm{Arg}\, R = \mathrm{Arg} a + 2\pi w$ holds: the sum of two principal arguments leaves $(-\pi, \pi]$ through $+\pi$ exactly when both are positive and the product has nonpositive argument, and through $-\pi$ exactly when both are negative and the product has nonnegative argument, while arguments of opposite sign cross no branch. At termination $a = \prod_{ij} R_{ij} = \prod_e |z_e|^2 > 0$, whose argument is 0, so $w = C$. All steps are exact on rationals. ■

Lemma 5. If $\tilde{C} \in \mathbb{R}$ satisfies $|\tilde{C} - C| < 1/2$ for some $C \in \mathbb{Z}$, then $\lfloor \tilde{C} + 1/2 \rfloor = C$.

Proof. From $C - 1/2 < \tilde{C} < C + 1/2$ one gets $C < \tilde{C} + 1/2 < C + 1$, so the floor equals C. ■

5. A perturbation certificate

Theorem 6. Suppose $\hat{z} = z(\hat{X})$ has nonvanishing links and plaquettes $\hat{R} = a + ib \notin (-\infty, 0]$. Put

$$\rho = \min_e \max(|\mathrm{Re}\hat{z}_e|, |\mathrm{Im}\hat{z}_e|) > 0, \qquad \beta_{ij} = \begin{cases} 1, & a \ge 0 \\ \dfrac{|b|}{|a| + |b|}, & a < 0 \end{cases}, \qquad \beta = \min_{ij} \beta_{ij}, \qquad d = \frac{2\kappa\varepsilon}{\hat{S}}.$$

If $\beta\rho - 8d > 0$, then every W' with $\| W' - \widehat{W} \|_1 \le \varepsilon$ has all sites admissible and $C(W') = C(\widehat{W})$.

Proof. First $\rho \le \min_e |\hat{z}_e|$. If $a \ge 0$ then $|\hat{P}_{ij} + 1| \ge \sqrt{2} \ge \beta_{ij}$; if $a < 0$, with $\varrho = \sqrt{a^2 + b^2}$, $|\hat{P}_{ij} + 1|^2 - b^2/\varrho^2 = (1 + a/\varrho)^2 \ge 0$, so $|\hat{P}_{ij} + 1| \ge |b|/\varrho \ge \beta_{ij}$. Thus β lower-bounds the distance of the observed unit plaquettes from -1. For any admissible W', Lemma 2 gives $\max_e |z'_e - \hat{z}_e| \le d$. Along $z_e(s) = \hat{z}_e + s(z'_e - \hat{z}_e)$, $s \in [0,1]$, one has $|z_e(s)| \ge \rho - d > 0$ since $8d < \beta\rho \le \rho$, and

$$\left| \frac{z_e(s)}{|z_e(s)|} - \frac{\hat{z}_e}{|\hat{z}_e|} \right| \le \frac{2|z_e(s) - \hat{z}_e|}{|\hat{z}_e|} \le \frac{2d}{\rho}.$$

Telescoping over the four unit factors of a plaquette gives $\left|P_{ij}(s) - \hat{P}_{ij}\right| \le 8d/\rho$, hence $\left|P_{ij}(s) + 1\right| \ge \beta - 8d/\rho > 0$. No link vanishes and no plaquette enters $(-\infty, 0]$ along the path, so by Lemma 3 C is constant in s, giving $C(W') = C\left(\widehat{W}\right)$; Lemma 4 computes it exactly. ■

6. Scale separation and non-identifiability

Lemma 7. Let $\mathcal{D}$ depend on its matrix argument only through X, so $\mathcal{D}(sW) = \mathcal{D}(W)$ for $s > 0$, and let $b \in \{0,1\}$ be determined by a finite history of values of $\mathcal{D}$. If a functional Q satisfies $Q(sL) = sQ(L)$ for all $s > 0$, then $Y := bQ$ satisfies $Y(sL) = sY(L)$. Moreover, for a real random variable L and $0 < \alpha < 1$, the quantile $q_\alpha(L) = \inf\{x: \mathbb{P}(L \le x) \ge \alpha\}$ satisfies $q_\alpha(sL) = sq_\alpha(L)$, and if $\int_\alpha^1 q_u \, du < \infty$ then $\mathrm{ES}_\alpha(L) = (1-\alpha)^{-1} \int_\alpha^1 q_u \, (L) \, du$ satisfies $\mathrm{ES}_\alpha(sL) = s \, \mathrm{ES}_\alpha(L)$.

Proof. A common rescaling leaves X, hence the history of $\mathcal{D}$ and thus b, unchanged, so $Y(sL) = bQ(sL) = sbQ(L) = sY(L)$. Since $\mathbb{P}(sL \le x) = \mathbb{P}(L \le x/s)$, the set defining the infimum scales by s, giving $q_\alpha(sL) = sq_\alpha(L)$; substitution into the integral gives the statement for ES_α. ■

Proposition 8. If $\mathcal{D}(x_1) = \mathcal{D}(x_2)$ while $Q(x_1) \ne Q(x_2)$, then no function of $\mathcal{D}$ alone equals Q at both points, and for any common value a, $\max(|Q(x_1) - a|, |Q(x_2) - a|) \ge 1/2 \, |Q(x_1) - Q(x_2)|$. In particular, if $\mathcal{D}(sx) = \mathcal{D}(x)$ for some $s > 0$, $s \ne 1$, and $Q(sx) = sQ(x)$ with $Q(x) \ne 0$, then $\mathcal{D}$ does not determine Q.

Proof. A function f with $f\left(\mathcal{D}(x_1)\right) = Q(x_1)$ and $f\left(\mathcal{D}(x_2)\right) = Q(x_2)$ has equal arguments but unequal values, a contradiction. The triangle inequality $|Q(x_1) - Q(x_2)| \le |Q(x_1) - a| + |a - Q(x_2)|$ forces one summand to be at least half the gap. In the homogeneous case $Q(sx) = sQ(x) \ne Q(x)$. ■

Proposition 9. If a connected family of configurations maps under a continuous embedding that avoids vanishing links and plaquettes in $(-\infty, 0]$, then C is constant on the family. For the baseline of Definition 2 with $\rho = \beta = 1$ and nonnegative entries summing to one, every $\kappa < 1/8$ yields C equal to the baseline charge for all such X.

Proof. C is continuous and integer-valued on the connected image, hence constant by Lemma 3. For the stated baseline each coordinate moves by at most κ, since the nonnegative entries sum to one; with reference distances $\rho = \beta = 1$ and raw bound $d = \kappa$, the argument of Theorem 6 applies whenever $8\kappa < 1$, keeping every such X in the baseline charge. ■

7. A conditional-expectation identity

Proposition 10. Let $\mathbb{E}[Y^2] < \infty$, $m = \mathbb{E}[Y \mid Z]$, $m_+ = \mathbb{E}[Y \mid Z, C]$. Then

$$\mathbb{E}[(Y - m)^2] - \mathbb{E}[(Y - m_+)^2] = \mathbb{E}[(m_+ - m)^2] \ge 0,$$

with strict inequality iff $\mathbb{P}(m_+ \neq m) > 0$. If C is $\sigma(V,Z)$-measurable, then $\mathbb{E}[Y \mid V,Z,C] = \mathbb{E}[Y \mid V,Z]$, so adjoining C to (V,Z) gives no reduction.

Proof. Write $Y - m = (Y - m_+) + (m_+ - m)$. The cross term has zero expectation, since $\mathbb{E}[Y - m_+ \mid Z,C] = 0$ and $m_+ - m$ is $\sigma(Z,C)$-measurable; the identity and strictness criterion follow. If $C = f(V,Z)$ then $\sigma(V,Z,C) = \sigma(V,Z)$, so the conditional expectations and their L^2 risks coincide. ■

8. The causal detection automaton

Theorem 11. In Definition 4, starting from BOOT: (i) output 1 first occurs only after $\geq h_B$ inputs of type S followed by h_A consecutive C; (ii) a return to output 0 from ARMED requires h_R consecutive S with elapsed time $\geq D$; (iii) any $\perp$ leaves ARMED in ARMED; (iv) an input stream consisting only of C never produces output 1. With fixed parameters and inputs ordered by strictly increasing time, the output at each step is a function of the inputs received up to that step, and equal prefixes give equal outputs.

Proof. BOOT exits only on its h_B threshold; READY reaches ARMED only on its h_A threshold, and any S or $\perp$ resets that counter; ARMED exits only on h_R consecutive S with the dwell test; the $\perp$ rule fixes ARMED and resets counters elsewhere; a C-only stream never meets the S threshold of BOOT, so it never leaves BOOT. Each transition reads only the current state, the fixed parameters, and the current input; induction on the step index gives causality and reproducibility. ■

Proposition 12. Let $(E_t)_{t=1}^{T}$ satisfy $\mathbb{P}(E_t \mid \mathcal{F}_{t-1}) \leq p$ almost surely for all t, where $\mathcal{F}_{t-1}$ is the history. For $h \leq T$, the probability that h consecutive E_t occur is at most $\min\{1, (T - h + 1)p^h\}$.

Proof. Such an occurrence lies in a union of at most $T - h + 1$ length-h runs; repeated conditioning bounds the probability of each run by p^h, and the union bound concludes. ■

Proposition 13. Suppose the automaton is in state READY before a time τ; every input at or after τ is admissible of type C; inputs arrive with gaps $\leq \Delta$ (the first by $\tau + \Delta$); each is processed within ℓ of arrival; and no freshness or gap condition fails. Then output 1 occurs by $\tau + h_A\Delta + \ell$, and for any distinguished time T^* the lead is at least $T^* - \tau - h_A\Delta - \ell$.

Proof. The first qualifying input is taken by $\tau + \Delta$; by induction on the gap bound the h_A-th is taken by $\tau + h_A\Delta$ and processed by $\tau + h_A\Delta + \ell$. That run of h_A admissible C inputs sends READY to ARMED at that step by Theorem 11, giving output 1. Subtracting the bound from T^* gives the lead. ■

9. A continuum functional: background

The following records the continuum origin of the integer invariant. It is included for context only; no result of Sections 2–8 depends on it.

Lemma 14. Let M be an oriented closed 3-manifold, A a fixed smooth matrix-valued 1-form in a fixed trivialization, and $k \in \mathbb{R}$. For

$$\mathcal{S}[A] = \frac{k}{4\pi} \int_M \operatorname{tr}\,(A \wedge dA + 2/3\ A \wedge A \wedge A),$$

if A, the orientation, and M do not depend on a metric g, then $\delta\mathcal{S}/\delta g_{\mu\nu} = 0$; any tensor defined as a nonzero density multiple of this variation vanishes.

Proof. The operations d, $\wedge$, tr, and integration of a top-degree form over an oriented manifold require no metric. For any smooth family $g(\epsilon)$ the integrand and the domain are fixed, so $\mathcal{S}$ is constant in ϵ and $d\mathcal{S}/d\epsilon = 0$; as the family is arbitrary, the functional derivative vanishes identically. ■

10. Distributed read-out

Lemma 15. Let ρ_{AB} be a state on $\mathcal{H}_A \otimes \mathcal{H}_B$ and $\{K_r\}$ operators on $\mathcal{H}_A$ with $\sum_r K_r^\dagger K_r = I$. Then $\sigma = \sum_r (K_r \otimes I)\ \rho_{AB}\ \left(K_r^\dagger \otimes I\right)$ has the same B-marginal as ρ_{AB}.

Proof. For any observable B on $\mathcal{H}_B$,

$$\operatorname{tr}[\sigma\,(I \otimes B)] = \sum_r \operatorname{tr}\left[(K_r \otimes I)\rho_{AB}\left(K_r^\dagger \otimes I\right)(I \otimes B)\right] = \operatorname{tr}\left[\rho_{AB}\left(\sum_r K_r^\dagger K_r \otimes B\right)\right] = \operatorname{tr}[\rho_{AB}(I \otimes B)],$$

using cyclicity of the trace and $[K_r \otimes I,\ I \otimes B] = 0$. As B is arbitrary, the marginals coincide. ■

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